

Generative AI

An overview of how it works, key examples, and real-world uses

What Is Generative AI?

Generative AI refers to artificial intelligence systems that create new content — text, images, audio, video, or code — rather than simply analyzing or classifying existing data. These models learn patterns, structures, and styles from large datasets, then use that knowledge to generate original outputs in response to a prompt.

How It Works

Most generative AI systems are built on deep learning architectures, particularly **transformers** for text and code, and **diffusion models** for images and video. During training, the model is exposed to massive amounts of data and learns to predict the most likely next piece of content (e.g. the next word or pixel pattern). Once trained, it can generate new, coherent content by sampling from what it has learned — guided by a user's prompt.

Common Examples	Real-World Applications
• Text: ChatGPT, Claude, Gemini • Images: Midjourney, DALL·E, Stable Diffusion • Audio/Music: Suno, ElevenLabs • Video: Sora, Runway • Code: GitHub Copilot, Claude Code	• Drafting emails, articles & reports • Software development & debugging • Marketing copy & design mockups • Customer support chatbots • Research summarization & data analysis • Personalized education & tutoring

Why It Matters

Generative AI is reshaping how people work and create by automating repetitive tasks, accelerating ideation, and lowering the barrier to producing professional-quality content. It is being adopted across industries — from software engineering and healthcare to marketing and education — as a tool that augments human creativity and productivity rather than replacing it entirely.

Key Considerations

Despite its benefits, generative AI raises important questions around accuracy (models can “hallucinate” incorrect information), bias inherited from training data, copyright and intellectual property, and the responsible disclosure of AI-generated content.